„A vision based reading system for color coded barcodes“

Erhard Schubert, Axel Schröder

Universität-Gesamthochschule Siegen

Zentrum für Sensorsysteme (ZESS)

Paul-Bonatz-Straße 9-11

57068 Siegen

Germany

ABSTRACT

Barcode systems are used to mark commodities, articles and products with price and article numbers. The advantage of the barcode systems is the safe and rapid availability of the information about the product. The size of the barcode depends on the used barcode system and the resolution of the barcode scanner. Nevertheless, there is a strong correlation between the information content and the length of the barcode. To increase the information content, new 2D-barcode systems like *CodaBlock* or *PDF-417* are introduced.

In this paper we present a different way to increase the information content of a barcode and we would like to introduce the color coded barcode. The new color coded barcode is created by offset printing of the three colored barcodes, each barcode with a different information. Therefore, three times more information content can be accommodated in the area of a black printed barcode. This kind of color coding is usable in case of the standard 1D- and 2D-barcodes.

We developed two reading devices for the color coded barcodes. First, there is a vision based system, consisting of a standard color camera and a PC-based color frame grabber. Omnidirectional barcode decoding is possible with this reading device. Second, a bi-directional handscanner was developed. Both systems use a color separation process to separate the color image of the barcodes into three independent grayscale images. In case of the handscanner the image consists of one line only. After the color separation the three grayscale barcodes can be decoded with standard image processing methods.

In principle, the color coded barcode can be used everywhere instead of the standard barcode. Typical applications with the color coded barcodes are found in the medicine technique, stock running and identification of electronic modules.

Keywords:
barcode, 2D-barcode, omnidirectional barcode decoding, color, color coding, image processing

0-8194-2039-5/96/$6.00

Downloaded From: http://proceedings.spiedigitallibrary.org/ on 07/17/2016 Terms of Use: http://spiedigitallibrary.org/ss/TermsOfUse.aspx

2 INTRODUCTION

The identification of commodities, articles, componenteries and products is a well-known problem of modern production and stock running. Medicine, specimens and canned blood have to be marked unambiguous in the medicine technique. In the retail trade goods have to be marked with price and article numbers. The barcode systems are an excellent solution for these or similar marking problems.

Any code number can be assigned to the object with barcodes. The main advantage of the barcode is that the code number can be read automatically with the barcode scanner. There are two different ways to read and to decode the barcode. First, the barcode is aligned to the scanner. In this case a bi-directional line scanner is needed. Second, the barcode is not aligned to the scanner. In this case we need an omnidirectional scanner to read the barcode. With an arrangement of mirrors and rotating mirrors it is possible to read the barcode in any orientation.

Admittedly, standard barcode systems can code a given number of digits only. The number of digits depends on the used type of barcode. The size of the barcode depends on the resolution of the barcode scanner and there is a strong correlation between the number of digits and the length of the barcode. Especially in the electro-technics and electronics the components become smaller and smaller. Therefore, the room for the identification label on the components becomes smaller and smaller, too. On the other hand, the number componenteries increases and because of this the length of the code numbers enlarges, too.

There are different ways to increase the density of the information in a given area:

- Every single information unit of the barcode can be reduced in size. This enlarges the expense of production and registration of the barcodes. The barcode printer and the reading devices rise into a high-price category.
- The standard barcode has the information along one direction only. By coding the information in both directions of the area it is possible to increase the density of the information considerably. This approach leads to the 2D-barcodes directly.
- Another option to increase the density of the information is to use a color coded barcode. Therefore, three times more information can be coded with the same size of a standard barcode. It is possible to use the color coding with standard barcodes and 2D-barcodes.

3 BARCODE SYSTEMS

3.1 Standard barcodes

The production of standard barcodes is easy. The reading and decoding devices are comparatively simple. The barcodes can be printed either on the object immediately or the barcodes can be pasted on as label. The size of the barcode depends on the kind of used characters (numerical, alphanumerical), the number of characters and the printer's resolution. Additionally the safety of reading is important. The safety of reading can be improved with a check sum or check digits. Therefore, the length of the barcode elongates, too. A great number of different barcode systems are created because of the different conditions and applications.

Examples:

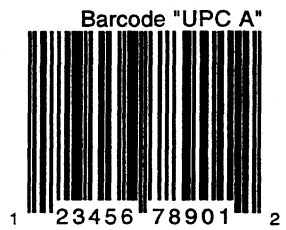


Figure 1

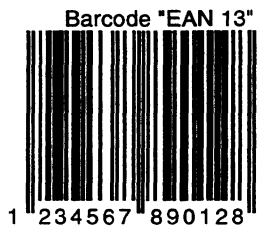


Figure 2



Figure 3

Downloaded From: http://proceedings.spiedigitallibrary.org/ on 07/17/2016 Terms of Use: http://spiedigitallibrary.org/ss/TermsOfUse.aspx

The UPC-code is used in the retail trade. The Codabar is used in the medicine technique and the code 2/5 interleaved, the code 128 and the code 39 are found in the industrial environment.

3.2 2D-barcodes

To increase the density of the information, 2D-barcodes are used more and more. As an example of the 2D-barcodes we have a look at the PDF-417. At first there is an example of the PDF-417. The 2D-barcode in Figure 4 includes this complete paragraph.



Figure 4: („To increase .... complete paragraph")

- **More information**
  The PDF-417 is characterized by the high density of information coupled with an excellent reading security. At the same size of a standard barcode more information can be included. Figure 5 and figure 6 show a comparison between the standard barcode Code 128 and the PDF-417 with the same size but different information.

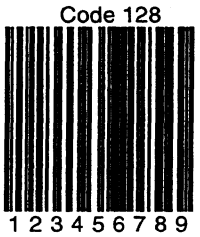


Figure 5: Standard-barcode



Figure 6: PDF-417 (13 times the information of the standard barcode from figure 5)

- **Character set**
  The complete ASCII-character set (PC437) can be coded with the PDF-417. Therefore, it is possible to code object description in plain text.

- **High security level**
  Many applications need a very high security of data transfer. The PDF-417 is created with a Reed-Solomon-Code with 9 security levels. This guarantees the readability of the barcode if the label is destroyed or covered. (See figure 7)

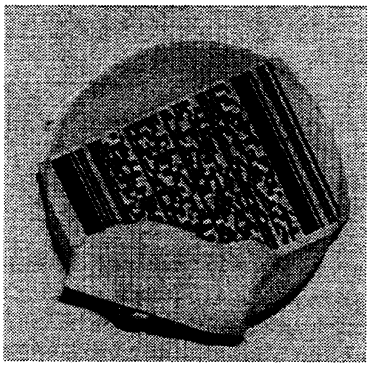


Figure 7: Partly covered PDF-417

Downloaded From: http://proceedings.spiedigitallibrary.org/ on 07/17/2016 Terms of Use: http://spiedigitallibrary.org/ss/TermsOfUse.aspx

- **Geometrical adjustment**
  The PDF-417 can be created with different ratio of height and length, so the PDF-417 is adjustable to different geometrical facts.

4 USING COLOR CODING TO INCREASE THE DENSITY OF THE INFORMATION

The first step to increase the density of the information is the usage of a 2D-barcode. With the introduction of a color coding an additional enlargement of the information density is possible.

The basic idea of the color coded barcodes originates in the video technique or the printing technique respectively. In these techniques the color separation is a main constituent. The visual information of a color image is divided at least into three color separations. In the video technique the color separations are called RGB channels. The printing technique uses the three color separations cyan, magenta and yellow. The black color separation of the printing technique is redundant. It includes this information only, which is the same in the cyan, magenta and yellow separation. Each of the three relevant color separations represents an image by its own with a different information. In the color image the three different images are combined to one image at the same area and they can be grabbed at the same time with a color sensitive sensor system.

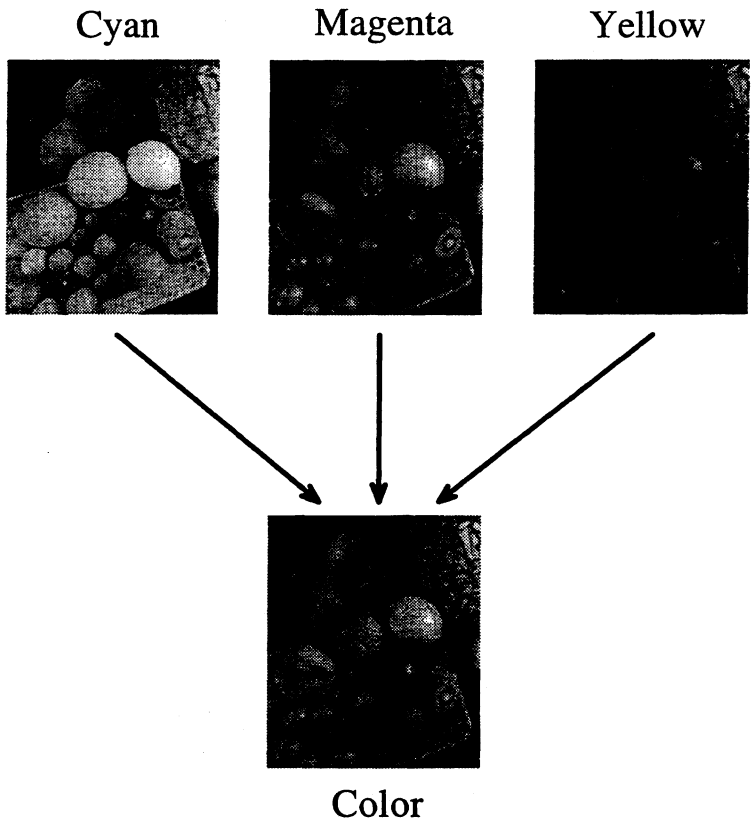


Figure 8: Color separation of a color image

Using this technique with the standard barcode or the 2D-barcodes, it is possible to code three barcodes with different information in the printing colors cyan, magenta and yellow and to print them one above the other at the same place. This color coded barcode can be read with a color sensitive sensor system. The three coded barcodes can be separated and decoded with standard methods of barcode decoding. This simple process allows to increase the information density by the factor of three. The color coding can be applied to both standard barcodes and 2D-barcodes

Downloaded From: http://proceedings.spiedigitallibrary.org/ on 07/17/2016 Terms of Use: http://spiedigitallibrary.org/ss/TermsOfUse.aspx

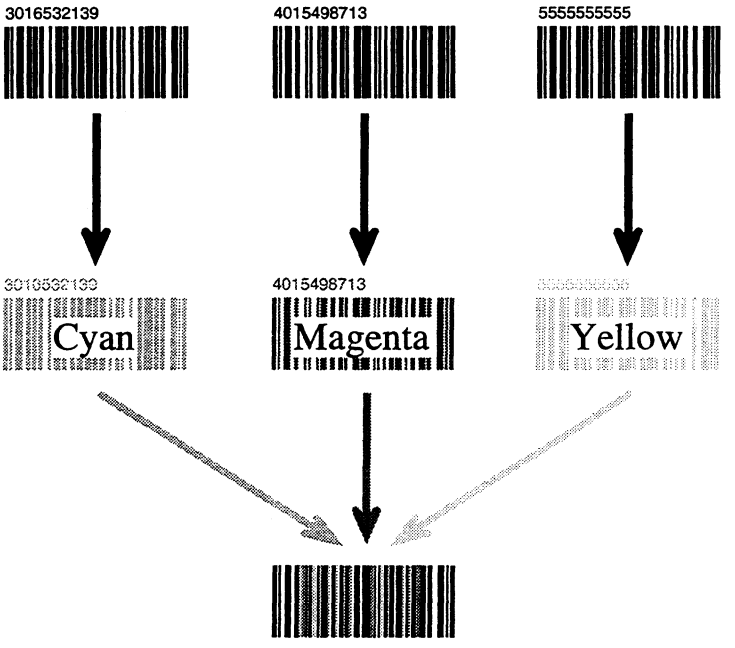


Figure 9: Example of color coded barcode

5 IMAGE PROCESSING

The first step of the image processing is the color separation. To extract the information from the color coded barcode the three printing colors have to be separated from the color image of the camera. The image of a color camera consists of three color separations, called the red, green and blue channel or RGB channels. There is no direct correlation between these RGB components and the printing colors cyan, magenta and yellow. Therefore, one of the main tasks of the image processing is the calculation of the printing color separations from the RGB channels.

The three color separations red, green and blue of the camera are linear independent. These three components put up a 3D-color space. The three printing inks also put up a 3D-color space. These two color spaces are complementary.

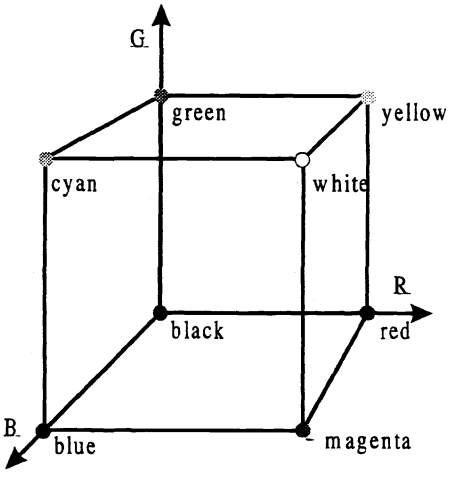


Figure 10: RGB-CMY color space

Downloaded From: http://proceedings.spiedigitallibrary.org/ on 07/17/2016 Terms of Use: http://spiedigitallibrary.org/ss/TermsOfUse.aspx

The main difference between these two color spaces is the kind of the color mixture. In case of the RGB color space the law of additive color mixture is valid while the law of absorptive color mixture is used in case of the CMY color space. Unfortunately there is no linear color space transformation between these two color spaces. Therefore, we use no mathematical color space transformation in our color separation algorithm. To realize a fast image processing, we simplify the complex color transformation into a 3D-LUT mapping.

5.1 3D-LUT calculation and mapping

There are eight colors mixed from the three printing inks cyan, magenta and yellow. First, there are the colors of the three printing inks themselves. Second, the colors red, green and blue appear. These colors result from the mixed printing of two inks. At last, there are the colors black and white. Black results from the mixed printing of all three printing inks and white results from the paper white if no ink is printed. To calculate the LUT, a test chart with all eight possible colors is printed and grabbed with the RGB camera. For each known color on the test chart, the RGB components are extracted. The resulting RGB values of the test colors are classified to one of eight regions in the RGB color space. The 3D-LUT is calculated with this classification result. The entries of the 3D-LUT contain three gray values for each combination of RGB values of the acquired color image. These three gray values correspond with the printing colors cyan, magenta and yellow. Figure 11 shows an example of the 3D-LUT with a RGB cluster of a color.

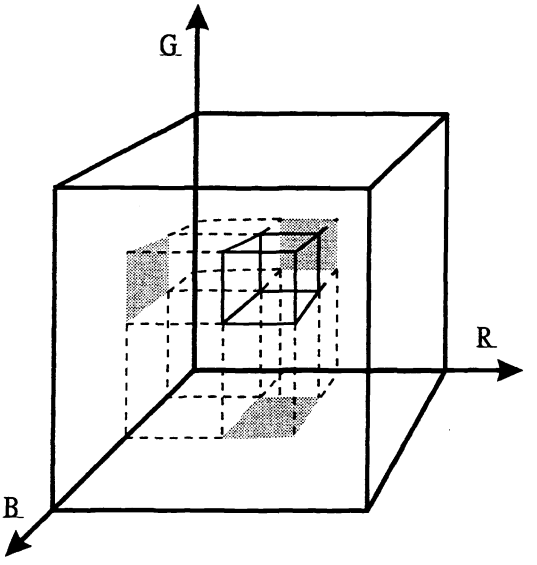


Figure 11: 3D-LUT with RGB cluster

The RGB values of each pixel of the grabbed image are used as addresses into the 3D-LUT and each combination of RGB values can be transformed unambiguously into a combination of printing inks. With the 3D-LUT mapping it is possible to reduce the color separation to only one processing step during the image acquisition. This makes real-time image processing possible. The color separation step leads directly to three grayscale images of the original barcodes.

5.2 Barcode decoding

The grayscale images of the barcodes are processed with standard algorithms of the grayscale image processing. First, the barcode is located in the image. The start and stop code of the barcodes are used for the position search. After the localization of the barcode the rotation and the size are determined. Therefore, the barcode reading is omnidirectional. There is no need to align the barcode with the camera. Depending on the rotation of the barcode, several scanlines are read out. After the binarizing of the profiles the information of the barcode is decoded. With an additional subpixel algorithm the performance of the system can be increased. The 2D-barcode are analyzed in this manner, too.

Downloaded From: http://proceedings.spiedigitallibrary.org/ on 07/17/2016 Terms of Use: http://spiedigitallibrary.org/ss/TermsOfUse.aspx

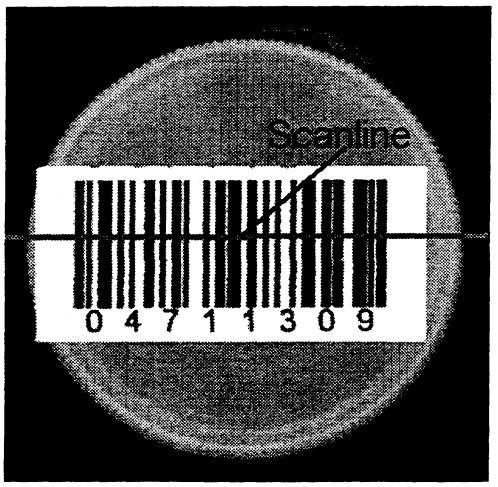


Figure 12: Standard-barcode with scanline after the color separation

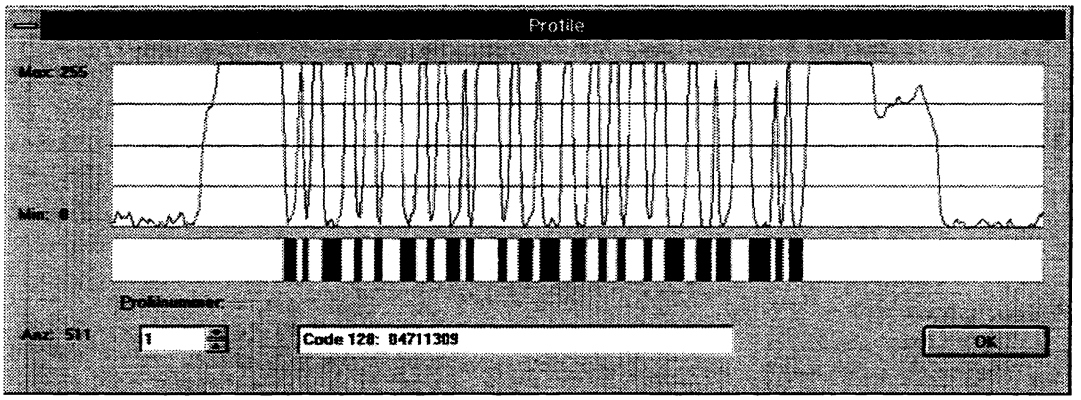


Figure 13: Gray Values along the scanline of Figure 12

6 READING DEVICES FOR COLOR CODED BARCODES

There are different reading devices for the color coded barcodes. These reading devices were developed from existing barcode scanners. Fundamentally, there are two groups of reading devices. First there are the bi-directional systems. In this case the barcode must be aligned to the scanner. The barcode is decoded perpendicular to the barcode lines in both directions, forward or backward. In the other case the barcode dose not have to be aligned to the scanner. These scanner systems are called omnidirectional scanners.

6.1 Bi-directional scanners

1.) Scanners with color selective sensor

Scanners with color selective sensor elements use CCD line sensors with RGB filters. They use a white light source as illumination. Before the barcode decoding the RGB color information of each line pixel must be converted into the CMY color space using the 3D-LUT mapping discussed above. The resulting grayscale profiles are decoded to obtain the barcode information.

Downloaded From: http://proceedings.spiedigitallibrary.org/ on 07/17/2016 Terms of Use: http://spiedigitallibrary.org/ss/TermsOfUse.aspx



Figure 14: Handscanner for color coded barcodes with RGB line sensor

2.) Scanner with selective illumination

Scanners with a selective illumination work with three illuminations with different spectral characteristics. The illuminations are used in time multiplex. The spectral characteristics are adapted to the used printing inks. The software color separation is not needed because the color separation is performed by the color selective illumination. A CCD line sensor is used as sensor element. The grayscale profiles are decoded with standard algorithms.

6.2 Omnidirectional scanners

A color CCD-camera in combination with a PC-based image processing hardware is used for the omnidirectional reading of color coded barcodes. With the color and image processing algorithms the barcodes can be decoded in all directions.

7 APPLICATIONS

Color coded barcodes can be used everywhere instead of standard barcodes. The main advantage is the automatic recognition of all barcode systems, the standard barcodes, the color coded barcodes and the 2D barcodes. Particularly the very high security level of the PDF-417 gives advantage even in a hard industrial environment.

7.1 Description and characterization of commodities in the production

Color coded barcodes can be used in the modern production process to mark different commodities and articles with all necessary information. Using the color coded PDF-417 it is possible to mark the objects with machine readable plain text. No database is needed.

7.2 Description of electronic components

Today service and a good support are an important decision reason. Using color coded barcodes and color coded 2D-barcodes electronic components and devices can be equipped with complete installation instructions. Each device is labeled with the valid instructions. Handbooks or handbook updates are unnecessarily. On the other hand the instructions are not accessible for nonprofessionals.

Downloaded From: http://proceedings.spiedigitallibrary.org/ on 07/17/2016 Terms of Use: http://spiedigitallibrary.org/ss/TermsOfUse.aspx

7.3 Shipping tickets and order forms

Shipping tickets and order forms can be provided with customer's information, delivery conditions, security references, etc. The whole information is machine readable and directly attachable to the objects.

7.4 Good description in storage

In the field of stock running and dispatching goods can be provided with machine readable labels. The staff gets all information at different terminals. With small hand devices the information is available even on a great site, for example a container terminal, without a connection to a central database.

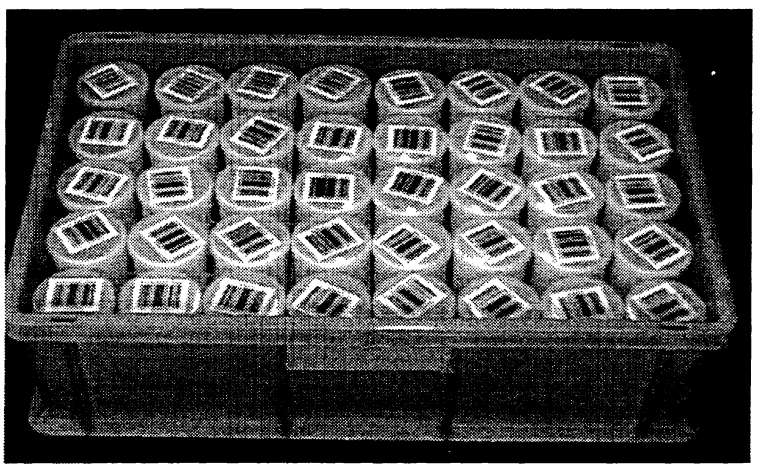


Figure 15: Box with labeled bottles

7.5 Description of medicine

Medicine and medical specimens can be labeled on a small area with important information about storing, transportation and use. Laboratories, pharmacies and transport companies are able to read the information automatically, fast and safe.

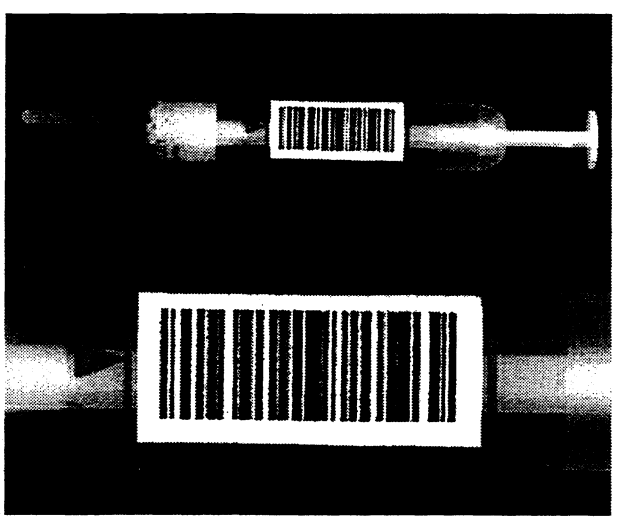


Figure 16: Monovette with colorbar Code128

Downloaded From: http://proceedings.spiedigitallibrary.org/ on 07/17/2016 Terms of Use: http://spiedigitallibrary.org/ss/TermsOfUse.aspx

8 CONCLUSIONS

The described method extends the existing barcode systems with a new and powerful approach, the color coded barcodes. With this method it is possible to increase the information density of the barcodes. The color coded barcode is created by offset printing of three colored barcodes. Therefore, three times more information can be accommodated with the size of a standard barcode. The color coding can be used with standard and with 2D-barcodes like the PDF-417. After a short introduction into the different barcode systems we described the method of the color coding of barcodes. Then we show the color separation process as a 3D-LUT mapping and the necessary image processing for omnidirectional barcode decoding. Finally we present the developed reading devices and some examples of applications.

9 ACKNOWLEDGEMENTS

The authors are grateful to the Center of Sensor Systems (Zentrum für Sensorsysteme, ZESS) at the University of Siegen for the financial and technical support.

10 REFERENCES

[1] Duda, R. O.; Hart, P. E. „Pattern classification and scene analysis", John Wiley & Son New York, 1973

[2] Rosenfeld, A.; Kak, A. C., „Digital Picture Processing, Vol. I + II", 2nd Edition, Academic Press, 1982

[3] Pratt, W. K., „Digital Picture Processing", John Wiley & Son New York, 1978

[4] Haberäcker, P., „Digitale Bildverarbeitung", 3. überarbeitete Auflage, Hanser Verlag, 1989

[5] Jähne, B., „Digitale Bildverarbeitung", 2. Auflage, Springer-Verlag, 1991

[6] Lang, H., „Farbmetrik und Farbfernsehn" R. Oldenbourg Verlag München Wien, 1978

[7] Wyszecki, G.; Stiles, W. S., „Color Science: Concepts and Methods, Qantitative Data and Formulae", John Wiley & Son New York, 1982

[8] Symbol Technologies Inc. and AIM Europe, „Uniform Symbology Specification - PDF417", 1994

[9] Hansen, H.-G.; Lenk, B.: „Codier Technik - Der Schlüssel zum Strichcode - ", Ident Verlag 1989

Downloaded From: http://proceedings.spiedigitallibrary.org/ on 07/17/2016 Terms of Use: http://spiedigitallibrary.org/ss/TermsOfUse.aspx